A-A-54374
12 April 1991

COMMERCIAL ITEM DESCRIPTION

CURING UNIT, DENTURE, ONE STAGE

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-C-37790 which is cancelled.

This commercial item description covers a one stage, automatically-controlled processing unit for curing dental acrylic resins used in the fabrication of dentures and other dental restoration and appliances.

Salient characteristics:

Fire and casualty hazards. The contractor shall submit proof to the contracting agency that the thermoswitch, on-off switch, strain relief, thermal cut-off, pilot light, wires, cord set, timer motor and timer switch he will supply under this specification conforms to the applicable requirements of the Underwriters Laboratory Inc. Standard UL 1262 for laboratory equipment. The label or listing of the Underwriters Laboratory Inc. may be accepted as evidence that they conform to this requirement.

Material.

Nickel-copper alloy (Monel). The nickel-copper alloy shall be in accordance with ASTM Designation E76 for composition only.

Corrosion-resistant steel. Class 3, type 302, 303 or 304 austenitic stainless steel and Class 4, type 430 martensitic stainless steel shall be in accordance with ASTM Designation F899 for composition only.

Design and dimensions. The curing unit shall be a one-stage, automatically controlled unit. The unit shall consist of an enclosure housing, a tank, a tank cover with a suitable handle, an electrical heated element, a power switch, a pilot light, an electrical timer of synchronous motor type, a strut type thermostat with calibrated temperature setting dial, and a thermal protection cut-off device. The design and dimensions of the curing unit will be as shown in figure 1. The tank shall be removable and of such size as to accommodate two standard denture flasks held in a two flask press for

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curing purposes. All controls, switches and pilot lights shall be mounted on the front control panel. The unit shall operate on 115 ± 5 volts, 60 Hertz (Hz) single phase alternating current. The power consumption shall be 350 watts ± 10%.

Enclosure housing. The enclosure housing shall consist of U-shaped, wrap-around forming the two sides and back, a front control panel, a baffle plate separating the electrical controls section from the heating section, a top and bottom plate, a tubular shell for holding the water tank, and a bottom cover for the tubular shell. There shall be a minimum of one inch spacing between the tubular shell and the U-shaped wrap-around. The top and bottom plate shall be die-cast aluminum with recesses cast in for receiving the U-shaped wrap-around and the center baffle. The top and bottom plates shall have a circular opening for receiving the tubular section. The U-shaped wrap-around, the tubular shell and the front panel shall be fabricated of corrosion-resistant steel having the thickness of 0.031 inch nominal. The baffle plate shall be fabricated from cold rolled steel having a thickness of 0.0598 inch nominal. The bottom cover for the tubular shell shall be fabricated from aluminum having a thickness of 0.025 inch nominal. The control panel shall be mounted to the wrap-around with a minimum of four corrosion-resistant metal machine screws. The housing shall be furnished with four, rubber covered, metal feet, secured at each corner of the bottom plate.

Water tank. The water tank shall be a deep drawn seamless tank. The tank shall be fabricated from corrosion-resistant steel having a thickness of 0.025 inch nominal. The tank shall have an inside diameter of 6-13/32 ± 1/8 inches and an inside height of 7-5/8 ± 1/8 inches. The tank shall have a rolled bead at the top edge to provide finger grip removal. The tank shall have a telescopic fit with the tubular shell and shall make contact with the thermostat when placed in position. The bottom of the tank shall be ribbed for reinforcement.

Water tank cover. The water tank cover shall be fabricated from corrosion-resistant steel having a thickness of 0.025 inch nominal. The cover shall be of the dome shape with a reverse draw to provide a drip ledge inside the tank for the condensate to drip back into the tank, when the cover is in place over the tank. A suitable handle, of the same material, shall be welded to the cover for lifting of the cover.

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Heating element. The heating element shall be of the disk type. The element wire shall be nickel-chrome steel imbedded in compressed refractory material and enclosed in type 430 stainless steel or nickel-copper alloy sheath. The heating element shall have heat-resistant terminals and shall be connected by stranded copper wire with a silicone rubber, glass braid jacket.

Thermostat. The thermostat shall be a strut-type construction, not bimetallic, enclosed in a stainless steel case and shall be mounted on the tubular shell with controls extended to the front control panel. The sensor shall contact the tank at all times when the tank is in operating position in the tubular shell. The thermostat shall be readily removable for servicing.

Temperature dial. The temperature dial shall be calibrated from 100°F to boiling and permanently and legibly marked in increments of 20°F. A pointer knob or other suitable means shall be provided to facilitate the selection of the desired temperature level. A reference line, at either extreme end of rotation, shall be provided for aligning the calibrated temperature dial indicator with the thermostat.

Timer. The timer shall consist of 60 Hertz (Hz), 115 volt, single phase, synchronous motor and the necessary mechanism to turn the heating unit off when the time has elapsed. The range of timer shall be from 15 minutes to 23 hours. The timer dial shall have a minimum diameter 1-7/8 inches and permanently and legibly marked in 15 minute intervals. The timer shall be mounted on the baffle plate with a circular window for viewing the dial, and a knob for setting the timer, provided on the front control panel. The timer shall be provided with a screw type terminal block for connection into the circuit.

Power switch. The power switch shall be a double pole, rotary switch, "on-off-on-off", four positions, per 360°, 90° between positions. The switch shall be rated for 15 amperes at 125 volt, 60 Hz. The switch shall be furnished with a knob dial permanently and legibly marked at the four positions. Power shall be supplied to the heating element only when the timer is in operation and thermostat is calling for heat.

Pilot light. The curing unit shall be equipped with a pilot light which shall indicate only when the timer is in operation, even though the power switch is on. The pilot light assembly shall have a non-corrosive plastic shell and shall be supplied with a red plastic lens.

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Thermal protection. The curing unit shall be equipped with a thermal protection device that will effectively provide a current interruption in the event that the thermostat malfunctions. The device shall be mounted on the bottom cast cover plate at the center of the controls section, forward of the baffle.

Power cord. The curing unit shall be equipped with a 3-conductor power cord not less than 6 feet in length. The cord shall be No. 16 AWG or No. 18 AWG, type SPT-2 or SPT-3 thermoplastic insulated, type HSJ rubber jacketed, or type HSJO rubber jacketed. The cord shall be equipped with a three prong plug suitable for use with receptacles specified in ANSI C73.11. The power cord shall enter the side of the unit through a strain relief fitting. The cord shall be suitably slit back and stripped for attachment of terminals for connection to the power switch and to the timer ground connection.

Performance.

Dielectric voltage. The curing unit shall be capable of withstanding without failure the dielectric voltage withstand test specified in Underwriters Laboratories Inc. Standard UL 1262 for laboratory equipment.

Power cord. The curing unit power cord shall be capable of withstanding without failure the strain relief test specified in Underwriters Laboratories Inc. Standard UL 1262 for laboratory equipment.

Operational test. The curing unit shall operate satisfactorily when tested as specified.

Heating rate. The heating element shall be capable of providing a uniform water temperature increase of 2-1/2° ± 1/2°F per minute when tested as specified.

Temperature control. The thermostat shall be capable of maintaining the temperature of the water within ± 2 percent of the temperature dial setting when tested as specified.

Thermal protection. The thermal protection device shall interrupt the electrical circuit at 210° ± 10°F when tested as specified.

Finish. The entire inside and outside surfaces of the water tank and cover, and exposed surfaces of tubular shell, shall be finished with a No. 4 finish, after all surface

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imperfections have been removed by grinding. As an alternate, the interior surface of the water tank may be electropolished and the outside surface and cover may be a bright polish. The finish shall result in a uniform, bright, shiny surface which is readily cleaned. All other exposed surfaces of the unit shall be finished in accordance with standard commercial practice. All non-corrosion resistant materials shall be suitably treated to resist corrosion.

Service data. The contractor shall furnish, with each unit, two copies of a technical manual which contains complete instructions for installation, operation and maintenance and a list of component parts.

Identification markings. Each curing unit shall be permanently and legibly marked with the following instructions: name or registered trademark of the manufacturer, manufacturer's model number, voltage, frequency, current and serial number.

Workmanship. The curing unit shall be free from defects such as burrs, feathers, sharp edges, pits, coarse grind marks and cracks which detract from its appearance or impair its serviceability.

Unit. Each (EA). One curing unit, as specified, constitutes one unit. Complete and detailed instructions for use shall be supplied with each unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and test performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

First article inspection. First article inspection shall be performed on the curing unit(s) when a first article sample(s) is required. This inspection shall include examination and tests as specified.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection, examination, and test characteristics. The unit of product for inspection, examination, test purposes shall be one curing unit. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective) and sampling for tests (when applicable) shall be conducted at inspection level S-1. Acceptance number for tests shall be zero (0) for all samples, except Dielectric strength inspection level shall be 100 percent and any defect shall constitute rejection of the lot.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Chemical analysis. In the absence of contractor certification for the raw material, a chemical analysis shall be conducted. The chemical analysis shall be performed by reagent solutions, combustion methods, nonemission methods, or by spectrochemical analysis. Any ASTM method shall be acceptable.

Operational test. The curing unit shall be given an operational test by turning the power on and setting the thermostat at 200°F. Observe if all components, except thermal protection switch, is functioning properly.

Heating rate. The following procedure shall be followed: Place a denture flask press with two denture flasks in the tank of the curing unit. Add sufficient water to cover the two

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flasks. Set the timer to 90 minutes and the thermostat to boiling. Suspend a calibrated immersion type thermometer in the water. Cover the tank with the thermometer protruding thru. Record the temperature of the water. Turn the main power switch on. Monitor the time it takes for the water temperature to reach 200°F. Calculate the heating rate.

Temperature control. The tank shall be filled with water to a depth of 4 ± 1/2 inches. A calibrated immersion type thermometer shall be suspended in the water. The tank shall be covered with the thermometer protruding thru. The thermostat shall be tested at intervals of 20°F starting at 100°F. The water temperature shall be permitted to stabilize. The thermostat setting and thermometer reading shall be recorded at each interval.

Thermal protection. At the completion of the heating rate test, the water shall be removed from the tank leaving 1-1/2 ± 1/2 inches of water at the bottom of the tank. A thermometer or other suitable means of measuring temperature shall be suspended in the controls section in the area of the thermal protection switch. The thermostat shall be shunted out of the circuit. The power shall be turned on and the curing unit permitted to operate without thermostatic control, and the water shall be permitted to evaporate completely. The curing unit shall be permitted to operate until the temperature in the area of the thermal switch reaches 210° ± 10°F. The temperature at which the thermal switch interrupts the circuit shall be recorded.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the

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offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from the receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-597-7749	CURING UNIT, DENTURE, ONE STAGE, 115 Volt, 60 Hertz

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

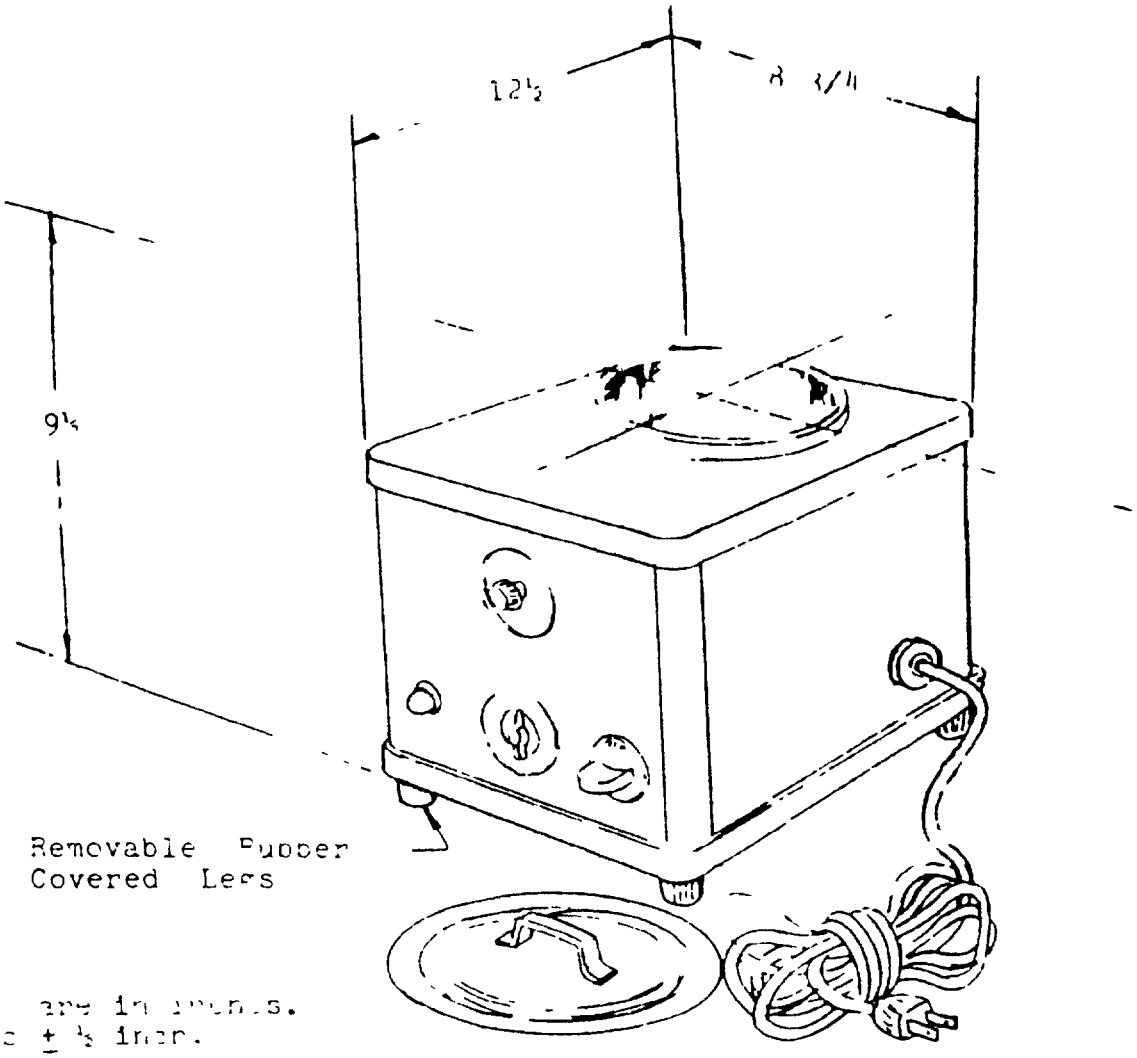
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2440
LOCATION: KD1/CURING1

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The illustration shown above is typical for style and design and is not intended to restrict or preclude the purchase of dental curing units which are otherwise in accordance with the requirements of this specification.

FIGURE 1. Curing unit, denture, one stage.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54374	2. DOCUMENT DATE (YYMMDD): 4-12-91

3. DOCUMENT TITLE: CURING UNIT, DENTURE, ONE STAGE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) / 7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial / (2) AUTOVON
Defense Personnel Support Center	(215)737-2870 / 444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSTE, 2800 S. 20th Street, Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466, Telephone (703) 756-2340, AUTOVON 289-2340

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